
2026 Chantilly Math Competition
Middle School Division - Team Round Solutions

1. (9 pts) How many subsets of $\{1, 2, 3, 4, 5, 6, 7\}$ have sum greater than 14?

Aryan Raj

Solution Note that $1 + 2 + \dots + 7 = \frac{1}{2} \cdot 7 \cdot 8 = 28 = 2(14)$.

Therefore, the set complement is a trivial bijection between subsets with sum less than 14 and subsets with sum greater than 14.

The subsets with sum exactly 14 are $\{7, 6, 1\}$, $\{7, 5, 2\}$, $\{7, 4, 3\}$, $\{6, 5, 3\}$, and the 4 complements, for 8 total.

Therefore, the answer is $\frac{2^7-8}{2} = \boxed{60}$.

2. (9 pts) Given that

$$((x^2 - 5x + 9)^2 - 5(x^2 - 5x + 9) + 9)^2 - 5((x^2 - 5x + 9)^2 - 5(x^2 - 5x + 9) + 9) + 9 = x$$

for some real number x , compute x .

Aryan Raj

Solution Let $f(x) = x^2 - 5x + 9$.

Then, we want $f(f(f(x))) = x$.

Since $f(x) - x = (x - 3)^2$, $f(x) \geq x$ with equality only at $x = 3$.

Therefore, $f(f(f(x))) \geq f(f(x)) \geq f(x) \geq x$ with equality only at $x = \boxed{3}$.

3. (10 pts) A palindrome is a positive integer that has the same value when its digits are reversed. Compute the smallest positive integer n such that n^2 is a palindrome with an even number of digits.

Aryan Raj

Solution Note that all palindromes with an even number of digits are divisible by 11.

Therefore, $11 \mid n$.

Since $11^2 = 121$, the possibility of a 2-digit palindrome is eliminated.

Now we consider 4-digit palindromes.

$$33^2 = 1089, 44^2 = 1936, 55^2 = 3025, 66^2 = 4356, 77^2 = 5929, 88^2 = 7744, 99^2 = 9801.$$

None of these are palindromes, so we consider 6-digit palindromes.

Let $n^2 = 100001a + 10010b + 1100c$ for integers $0 \leq a, b, c \leq 9$ and $a > 0$.

By considering the last digit of a square, $a \in \{1, 4, 5, 6, 9\}$.

When $a = 1$, n must be a multiple of 11 ending in 1 or 9 between $\sqrt{100000}$ and $\sqrt{200000}$.

$$319^2 = 101761, 341^2 = 116281, 429^2 = 184041.$$

When $a = 4$, n must be a multiple of 11 ending in 2 or 8 between $\sqrt{400000}$ and $\sqrt{500000}$.

$$638^2 = 407044, 682^2 = 465124.$$

When $a = 5$, n must be a multiple of 11 ending in 5 between $\sqrt{500000}$ and $\sqrt{600000}$.

$$715^2 = 511225.$$

When $a = 6$, n must be a multiple of 11 ending in 4 or 6 between $\sqrt{600000}$ and $\sqrt{700000}$.

$$814^2 = 662596, 836^2 = 698896.$$

Therefore, the answer is 836.

4. (10 pts) Shubham is at some uniformly random point (a, b) where a and b are both integers between -2 and 2 inclusive. Every minute, Shubham moves to a point with integer coordinates that is exactly $\sqrt{5}$ away from the point he is currently at. What is 100 times the expected value of the smallest possible number of minutes until Shubham is at $(0, 0)$?

Shubham Patel

Solution Refer to the following diagram, where F represents $(0, 0)$.

A	B	C	B	A
B	D	E	D	B
C	E	F	E	C
B	D	E	D	B
A	B	C	B	A

From every A Shubham needs ≥ 4 minutes.

From every B Shubham needs ≥ 1 minutes.

From every C Shubham needs ≥ 2 minutes.

From every D Shubham needs ≥ 2 minutes.

From every E Shubham needs ≥ 3 minutes.

From every F Shubham needs ≥ 0 minutes.

Therefore, the expected value of the smallest possible number of minutes is

$$\frac{4 \cdot 4 + 8 \cdot 1 + 4 \cdot 2 + 4 \cdot 2 + 4 \cdot 3 + 1 \cdot 0}{25} = \frac{52}{25},$$

so the answer is $100 \cdot \frac{52}{25} = \boxed{208}$.

5. (11 pts) Define $\tau(n)$ as the number of positive integer divisors of n .

Compute the sum of all integers n such that $1 \leq n \leq 100$ and $2\tau(\tau(n)) = \tau(n)$.

Aryan Raj

Solution $2\tau(x) = x$ is only true for $x \in \{8, 12\}$, because when $x > 16$, $2\tau(x) \leq 4\sqrt{x} < x$.

When $\tau(n) = 8$, the possible prime factorizations of n are pqr and p^3q .

When $n = pqr$, we get $2 \cdot 3 \cdot 5 + 2 \cdot 3 \cdot 7 + 2 \cdot 3 \cdot 11 + 2 \cdot 3 \cdot 13 + 2 \cdot 5 \cdot 7 = 286$.

When $n = p^3q$, we get $2^3 \cdot 3 + 2^3 \cdot 5 + 2^3 \cdot 7 + 2^3 \cdot 11 + 3^3 \cdot 2 = 262$.

When $\tau(n) = 12$, the possible prime factorizations of n are p^5q , p^3q^2 , and p^2qr .

We get $2^5 \cdot 3 + 2^3 \cdot 3^2 + 2^2 \cdot 3 \cdot 5 + 2^2 \cdot 3 \cdot 7 + 3^2 \cdot 2 \cdot 5 = 402$.

The answer is $286 + 262 + 402 = \boxed{950}$.

6. (11 pts) There are 6 cities, every pair of them is connected by a road, and it is impossible to switch roads without arriving at a city. It is snowing, so every hour, 1 of these roads is closed. After 2 hours, the probability that it is possible to start in some city and drive along all the non-closed roads, each exactly once, can be expressed as $\frac{a}{b}$ for relatively prime positive integers a and b . Compute $10a + b$.

Aryan Raj

Solution It is possible if and only if exactly 0 or 2 cities are connected by non-closed roads to an odd number of other cities.

Before any roads are closed, every city is connected to 5 other cities.

After 2 hours, we need the 2 closed roads to not have a common city, so that 4 cities will be connected to 4 other cities, and exactly 2 cities will be connected to 5 other cities.

There are $\binom{6}{2} - 1 = 14$ possibilities for the 2nd closed road.

Only $\binom{4}{2} = 6$ of these will be disjoint from the 1st closed road.

Therefore, the desired probability is $\frac{6}{14} = \frac{3}{7}$, so the answer is $10(3) + 7 = \boxed{37}$.

7. (12 pts) Compute the largest of the 5 distinct primes that are divisors of $2^{41} + 2^{67}$.

Aryan Raj

Solution 2 is 1 of the primes. Factoring out all 2s gives $1 + 2^{26}$.

$$1 + 2^{26} = (1 + 2^{13})^2 - (2^7)^2 = (1 - 2^7 + 2^{13})(1 + 2^7 + 2^{13}).$$

$$1 - 2^7 + 2^{13} = 1 - 128 + 8192 = 8065 = 5 \cdot 1613.$$

$$1 + 2^7 + 2^{13} = 1 + 128 + 8192 = 8321.$$

By checking the 12 primes up to $\sqrt{1613}$, or 37, we see that 1613 is prime.

Since there are 5 prime divisors of $2^{41} + 2^{67}$, but we only have 2, 5, and 1613 so far, we see that 8321 must be divisible by 2 distinct primes.

Since $2, 3, 5 \nmid 8321$ and $\frac{8321}{7} < 1613$, both primes that divide 8321 are less than 1613.

Therefore, the answer is $\boxed{1613}$.

8. (12 pts) Let $ABCD$ be a parallelogram such that $AB = 41$ and $BC = 82$. Let O be the intersection of AC and BD . Let T and I lie on AD and AC , respectively, so that they both lie on the internal angle bisector of $\angle ABC$. Given that $AC = 67$, compute the area of parallelogram $ABCD$ divided by the area of quadrilateral $DOIT$.

Aryan Raj

Solution For brevity, let $AB = a$, $BC = b$, and $AC = c$.

Furthermore, let $[X_1X_2 \dots X_n]$ denote the area of polygon $X_1X_2 \dots X_n$.

By Angle Bisector Theorem, $AI = \frac{ac}{a+b}$ and $CI = \frac{bc}{a+b}$.

It is easy to see that $\triangle AIT \sim \triangle CIB$, so

$$AT = AI \cdot \frac{CB}{CI} = \frac{ac}{a+b} \cdot \frac{a+b}{c} = a.$$

Therefore,

$$\frac{[AIT]}{[AOD]} = \frac{AI}{AO} \cdot \frac{AT}{AD} = \frac{2a}{a+b} \cdot \frac{a}{b} = \frac{2a^2}{ab+b^2}.$$

It suffices to compute

$$\frac{[ABCD]}{[DOIT]} = \frac{4[AOD]}{[AOD] - [AIT]} = \frac{4}{1 - \frac{[AIT]}{[AOD]}} = \frac{4}{1 - \frac{2a^2}{ab+b^2}} = \frac{4ab+4b^2}{ab+b^2-2a^2}.$$

Since $b = BC = 82 = 2 \cdot 41 = 2AB = 2a$,

$$\frac{4ab+4b^2}{ab+b^2-2a^2} = \frac{8a^2+16a^2}{2a^2+4a^2-2a^2} = \frac{24}{4} = \boxed{6}.$$

9. (13 pts) How many nonnegative integers have at most 6 digits, have no digit larger than 7, and contain the digit 6 immediately followed by the digit 7 somewhere inside it?

Aryan Raj

Solution 1 We use the Principle of Inclusion-Exclusion.

The answer will be

$$\#(\text{have } \geq 1 \text{ } 67) - \#(\text{have } \geq 2 \text{ } 67) + \#(\text{have } \geq 3 \text{ } 67).$$

When there is ≥ 1 67, there are $6 - 2(1) = 4$ slots left, so

$$\#(\text{have } \geq 1 \text{ } 67) = 5 \cdot 8^4.$$

When there is ≥ 2 67, there are $6 - 2(2) = 2$ slots left, so

$$\#(\text{have } \geq 2 \text{ } 67) = 6 \cdot 8^2.$$

When there is ≥ 3 67, there are $6 - 2(3) = 0$ slots left, so

$$\#(\text{have } \geq 3 \text{ } 67) = 1.$$

Therefore,

$$\#(\text{have } \geq 1 \text{ } 67) - \#(\text{have } \geq 2 \text{ } 67) + \#(\text{have } \geq 3 \text{ } 67) = 5 \cdot 8^4 - 6 \cdot 8^2 + 1 = \boxed{20097}.$$

Solution 2 This nonnegative integer is a list of 6 nonnegative integers between 0 and 7 inclusive. We use recursion.

Let $A(n)$ denote the number of lists of the first n digits not having a 6 immediately followed by a 7, but the n th digit is not 6.

Let $B(n)$ denote the number of lists of the first n digits not having a 6 immediately followed by a 7, but the n th digit is a 6.

We want $8^6 - (A(6) + B(6))$.

$A(n+1) = 7A(n) + 6B(n)$ because we either add anything but a 6 to a $A(n)$ list or add anything but a 6 or 7 to a $B(n)$ list.

$B(n+1) = A(n) + B(n)$ because we add a 6 to a $A(n)$ list or add a 6 to a $B(n)$ list.

Let $T(n) = A(n) + B(n)$, so we want $8^6 - T(6)$.

Then, adding both recurrences gives $T(n+1) = 8T(n) - B(n)$.

The second recurrence simplifies into $B(n+1) = T(n)$.

Substitution gives $T(n+1) = 8T(n) - T(n-1)$.

We know that $T(0) = 1$ and $T(1) = 8$, because you can't have a 6 followed by a 7 in a list shorter than 2 digits.

$$T(2) = 8 \cdot 8 - 1 = 63, T(3) = 8 \cdot 63 - 8 = 496, T(4) = 8 \cdot 496 - 63 = 3905,$$

$$T(5) = 8 \cdot 3905 - 496 = 30744, T(6) = 8 \cdot 30744 - 3905 = 242047$$

Therefore, $8^6 - T(6) = 262144 - 242047 = \boxed{20097}$.

10. (13 pts) All 4 faces of a tetrahedron have side lengths of 6, 7, and 7. Given that the volume of this tetrahedron can be expressed as $a\sqrt{b}$ where a and b are positive integers and b isn't divisible by the square of a prime, compute $a + 10b$.

Aryan Raj

Solution We see that the tetrahedron will have a pair of opposite edges with length 6, and the other 4 edges will have length 7.

Consider the line segment connecting the midpoints of both edges with length 6.

This line segment has length $\sqrt{7^2 - (\frac{6}{2})^2 - (\frac{6}{2})^2} = \sqrt{31}$.

The 2 planes going thru each edge of length 6 and that line thru the midpoints will divide the tetrahedron into equal quarters.

Each quarter is a tetrahedron with orthogonal side lengths of 3, 3, and $\sqrt{31}$.

Therefore, each quarter has volume $\frac{1}{6} \cdot 3 \cdot 3 \cdot \sqrt{31} = \frac{3}{2}\sqrt{31}$.

Multiplying by 4 gives the total volume of $6\sqrt{31}$.

Therefore, the answer is $(6) + 10(31) = \boxed{316}$.

11. (14 pts) Soham thinks of some positive integer n . How many distinct nonnegative integers could be the remainder when n^n is divided by 2025?

Aryan Raj

Solution The quantity of interest in the problem is $n^n \bmod 2025$.

Since $2025 = 25 \cdot 81$ and $\gcd(25, 81) = 1$, we combine $n^n \bmod 25$ and $n^n \bmod 81$ using the Chinese Remainder Theorem.

When $\gcd(n, 15) = 1$, n is coprime to both 25 and 81. Then $n^n \bmod 25$ can be any number coprime to 25 and $n^n \bmod 81$ can be any number coprime to 81. Hence, there are $\phi(25)\phi(81) = 20 \cdot 54 = 1080$ values.

When $\gcd(n, 15) = 3$, $81 \mid n^n$ and $n^n \bmod 25$ can be any number coprime to 25. The exception is $n = 3$, where $81 \nmid 3^3$. Hence, there are $\phi(25) + 1 = 20 + 1 = 21$ values.

When $\gcd(n, 15) = 5$, $25 \mid n^n$ and $n^n \bmod 81$ can be any number coprime to 81. Hence, there are $\phi(81) = 54$ values.

When $\gcd(n, 15) = 15$, $25 \mid n^n$ and $81 \mid n^n$. Hence, there is 1 value.

Summing over all 4 cases gives the answer of $1080 + 21 + 54 + 1 = \boxed{1156}$.

12. (14 pts) Compute the maximum possible value of

$$2025 \cdot \frac{x^4 \cdot 4^x}{2^{x \cdot 6} + 2x^6}$$

over all real numbers x .

Nischal Annapareddy

Solution By the Arithmetic Mean-Geometric Mean Inequality,

$$2^{x \cdot 6} + 2x^6 = 4^{3x} + x^6 + x^6 \geq 3\sqrt[3]{4^{3x}x^6x^6} = 3 \cdot 4^x \sqrt[3]{x^{12}} = 3 \cdot x^4 \cdot 4^x,$$

with equality at the real number x such that $64^x = x^6$.

Therefore,

$$2025 \cdot \frac{x^4 \cdot 4^x}{2^{x \cdot 6} + 2x^6} \leq 2025 \cdot \frac{1}{3} = \boxed{675}.$$

13. (15 pts) Jonathan has a deck of 52 cards. Each card has 1 number on it. For each of the numbers 1, 2, 3, and 4, there are 13 cards in the deck with that number on it. A group of 4 cards is called awesome if and only if no 2 cards in the group have the same number. Jonathan splits his deck of 52 cards into 13 disjoint groups of 4 cards. Given that the expected number of these 13 groups that are awesome can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n , compute $m + n$.

Riyan Nayak

Solution The probability of group of 4 cards being awesome is

$$\frac{4! \cdot 13^4}{52 \cdot 51 \cdot 50 \cdot 49} = \frac{2197}{20825}.$$

By Linearity of Expectation, the desired expectation is simply 13 times this probability.

This is $13 \cdot \frac{2197}{20825} = \frac{28561}{20825}$, for an answer of $28561 + 20825 = \boxed{49386}$.

14. (15 pts) Ray has an integer $t = 0$, and is facing north. Every minute, Ray turns t degrees clockwise, and if he is facing north after that turn, he increases t by 1. In how many minutes will $t = 360$?

Aryan Raj

Solution We claim that it takes $\frac{360}{\gcd(k, 360)}$ minutes to go from $t = k$ to $t = k + 1$.

Once $t = k$, after m minutes, mk degrees will have been turned.

The smallest integer m for which $360 \mid mk$ is $\frac{360}{\gcd(k, 360)}$, as desired.

It suffices to compute the sum of

$$\sum_{k=0}^{359} \frac{360}{\gcd(k, 360)}.$$

By Chinese Remainder Theorem, this is equal to

$$\left(\sum_{k=0}^7 \frac{8}{\gcd(k, 8)}\right) \left(\sum_{k=0}^8 \frac{9}{\gcd(k, 9)}\right) \left(\sum_{k=0}^4 \frac{5}{\gcd(k, 5)}\right).$$

The answer is $(1 \cdot 1 + 1 \cdot 2 + 2 \cdot 4 + 4 \cdot 8)(1 \cdot 1 + 2 \cdot 3 + 6 \cdot 9)(1 \cdot 1 + 4 \cdot 5) = \boxed{55083}$.

15. (16 pts) Jake draws a hexagon. All its angles are equal. All its side lengths are either 6 or 7. The sum of all distinct possible areas of this hexagon is $\frac{m}{n}\sqrt{3}$ where m and n are relatively prime positive integers. Compute $m + n$.

Aryan Raj

Solution Let the hexagon's side lengths in order be a, b, c, d, e, f .

Extending all sides gives 2 equilateral triangles, giving the equations

$$a + b + c = c + d + e = e + f + a \text{ and } b + c + d = d + e + f = f + a + b$$

.

If we have 3 6s in a row, then, WLOG, $a + b + c = 18$, so $c + d + e = 18$ and $e + f + a = 18$, so $a = b = c = d = e = f = 6$.

Similarly, if we have 3 7s in a row, then $a = b = c = d = e = f = 7$.

Therefore, assume we never have the same number 3 times in a row.

If we have 2 6s in a row, then the next number must be a 7, and, WLOG,

$$a + b + c = 6 + 6 + 7 = 19,$$

so $c + d + e = 7 + d + e = 19$, and $e + f + a = e + f + 6 = 19$, so $d + e = 12$ and $e + f = 13$, so $a = b = d = e = 6$ and $c = f = 7$.

Similarly, if we have 2 7s in a row, then WLOG, we have $a = b = d = e = 7$ and $c = f = 6$.

The only case where we don't have the same number twice in a row is, WLOG, $a = c = e = 6$ and $b = d = f = 7$.

Therefore, we have 5 distinct possible areas.

We can use our equilateral triangles from earlier, along with the formula that the area of an equilateral triangle with side length x is $\frac{1}{4}x^2\sqrt{3}$, to get the following:

$$a = b = c = d = e = f = 6 \text{ gives } 54\sqrt{3}$$

$$a = b = c = d = e = f = 7 \text{ gives } \frac{147}{2}\sqrt{3}$$

$$a = b = d = e = 6 \text{ and } c = f = 7 \text{ gives } 60\sqrt{3}$$

$$a = b = d = e = 7 \text{ and } c = f = 6 \text{ gives } \frac{133}{2}\sqrt{3}$$

$$a = c = e = 6 \text{ and } b = d = f = 7 \text{ gives } \frac{253}{4}\sqrt{3}$$

Since $54 + \frac{147}{2} + 60 + \frac{133}{2} + \frac{253}{4} = \frac{1269}{4}$, the answer is $1269 + 4 = \boxed{1273}$.

16. (16 pts) For all non-negative integers d , let P_d be the unique polynomial of degree d such that

$$P_d(2^k) = 2^{k+1}$$

for all non-negative integers k less than d , and $P_d(2^d) = 1$.

Given that

$$P_0(5) + P_1(4) + P_2(3) + P_3(2) + P_4(1) + P_5(0)$$

can be expressed as $\frac{a}{b}$ for relatively prime positive integers a and b , compute $a + b$.

Aryan Raj

Solution Clearly, $P_3(2) = 4$ and $P_4(1) = 2$ by definition.

Since P_0 is constant and $P_0(2^0) = 1$, we have $P_0(5) = 1$.

Since P_1 is linear and $P_1(2^0) = 2^1$ and $P_1(2^1) = 1$, we get $P_1(x) = 3 - x$, so $P_1(4) = 3 - 4 = -1$.

Therefore, the desired sum is $6 + P_2(3) + P_5(0)$.

Let $Q_d(x) = P_d(x) - 2x$.

This has roots at $x = 2^k$ for all non-negative integers k less than d .

Therefore,

$$Q_d(x) = a(x - 2^0) \cdot \dots \cdot (x - 2^{d-1}),$$

for some a .

Since $Q_d(2^d) = P_d(2^d) - 2(2^d) = 1 - 2^{d+1}$, we get

$$a = \frac{1 - 2^{d+1}}{(2^d - 2^0) \cdot \dots \cdot (2^d - 2^{d-1})}.$$

Therefore,

$$Q_2(3) = \frac{1 - 2^3}{(2^2 - 2^0)(2^2 - 2^1)} \cdot (3 - 2^0)(3 - 2^1) = -\frac{7}{3},$$

so $P_2(3) = -\frac{7}{3} + 2(3) = \frac{11}{3}$.

Similarly,

$$Q_5(0) = \frac{1 - 2^6}{(2^5 - 2^0) \cdot \dots \cdot (2^5 - 2^4)} \cdot (-2^0) \cdot \dots \cdot (-2^4).$$

Using $\frac{-2^c}{2^d - 2^c} = \frac{1}{1 - 2^{d-c}}$, we get

$$Q_5(0) = \frac{1 - 2^6}{(1 - 2^1) \cdot \dots \cdot (1 - 2^5)} = \frac{63}{1 \cdot 3 \cdot 7 \cdot 15 \cdot 31} = \frac{1}{155}.$$

Thus, $P_5(0) = \frac{1}{155} + 2(0) = \frac{1}{155}$.

Therefore, the desired sum is $6 + \frac{11}{3} + \frac{1}{155} = \frac{4498}{465}$, so the answer is $4498 + 465 = \boxed{4963}$.

- TB. (Tiebreak) Ryan has a random number generator with 2 integer parameters $1 = m < M = 67$. While $M - m > 1$, the random number generator picks some integer between m and M exclusive, and Ryan changes either m or M , each option with probability $\frac{1}{2}$, to that integer. Estimate the integer nearest to 1 billion times the probability that $m = 41$ when $M - m = 1$.

Aryan Raj

Solution Let $A(x, y)$ be the answer when starting with $m = x$ and $M = y$.

When $y - x = 1$, $A(x, y) = 1$ if $x = 41$ and $A(x, y) = 0$ otherwise.

Otherwise,

$$A(x, y) = \frac{1}{y - x - 1} \sum_{k=x+1}^{y-1} \frac{A(x, k) + A(k, y)}{2}.$$

Coding up this recursion gives

$$A(1, 67) = \frac{424689107887844190802240558129596195}{42535295865117307932921825928971026432} \approx .009984392943558356.$$

Therefore, the integer nearest to $10^9 A(1, 67)$ is 9984393.